



ENSEA

Beyond Engineering

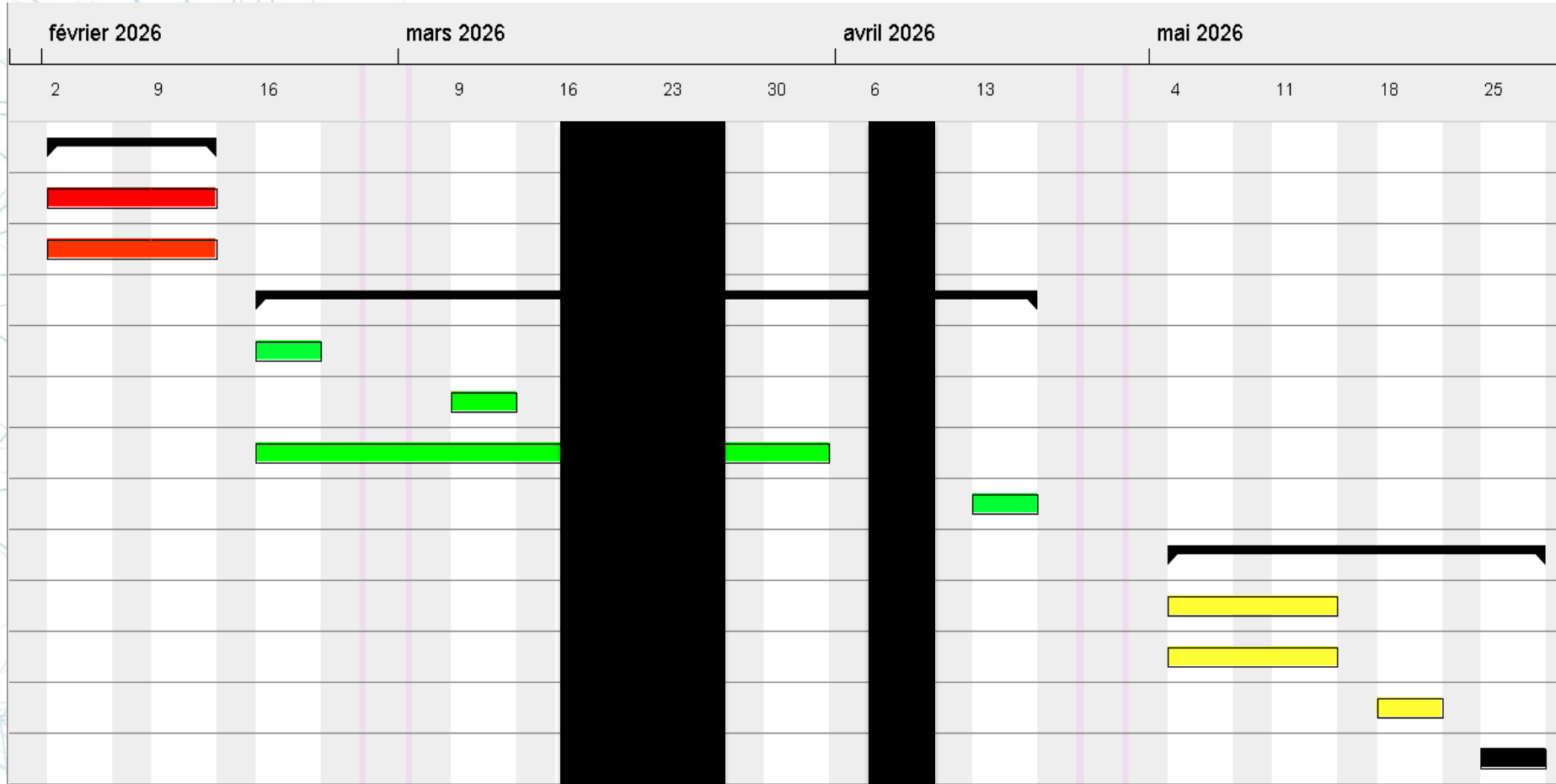
Project S6: Mid-term review

5th week

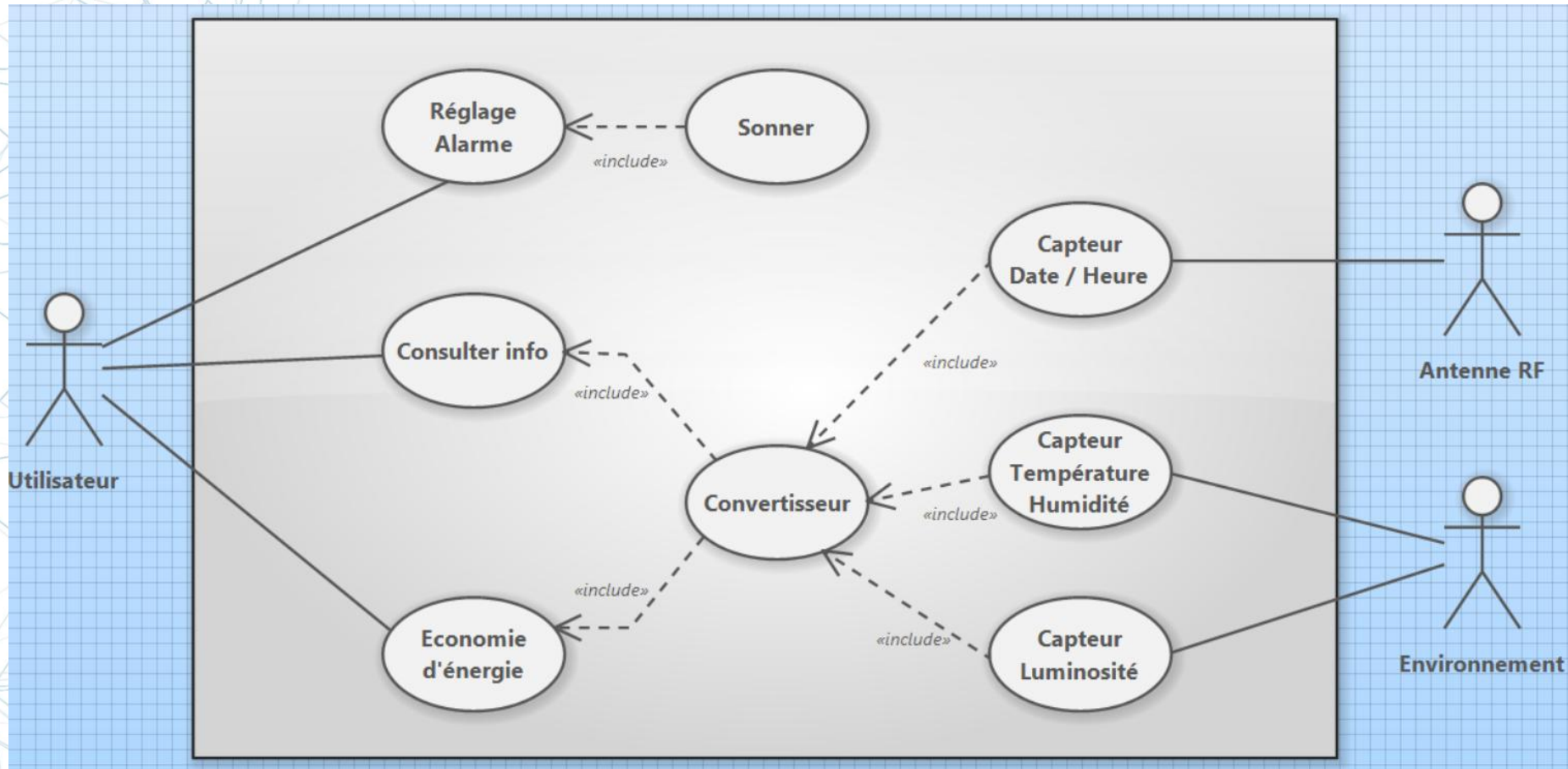
Forecast Schedule

Gantt project			
	Nom	Date de début	Date de fin
▼	Phase 1 : Analyse & Définition	02/02/2026	09/02/2026
	Analyse Fonctionnelle	02/02/2026	09/02/2026
	Recherche et choix des composants	02/02/2026	09/02/2026
▼	Phase 2 : Réalisation Technique	16/02/2026	13/04/2026
	Commande matériel	16/02/2026	16/02/2026
	Réception matériel	09/03/2026	09/03/2026
	Conception PCB KiCAD (Alimentation, Acquisition, Affichage)	16/02/2026	30/03/2026
	Revue de projet Mi-parcours	13/04/2026	13/04/2026
▼	Phase 3 : Intégration & Clôture	04/05/2026	25/05/2026
	Programmation composants STM32	04/05/2026	11/05/2026
	Soudure des composants (après fabrication PCB pendant les vacance...	04/05/2026	11/05/2026
	Assemblage final et test du mode économie d'énergie.	18/05/2026	18/05/2026
	Finalisation du rapport + Soutenance.	25/05/2026	25/05/2026

3 phases: **Definition**, **Conception**, **Fabrication**

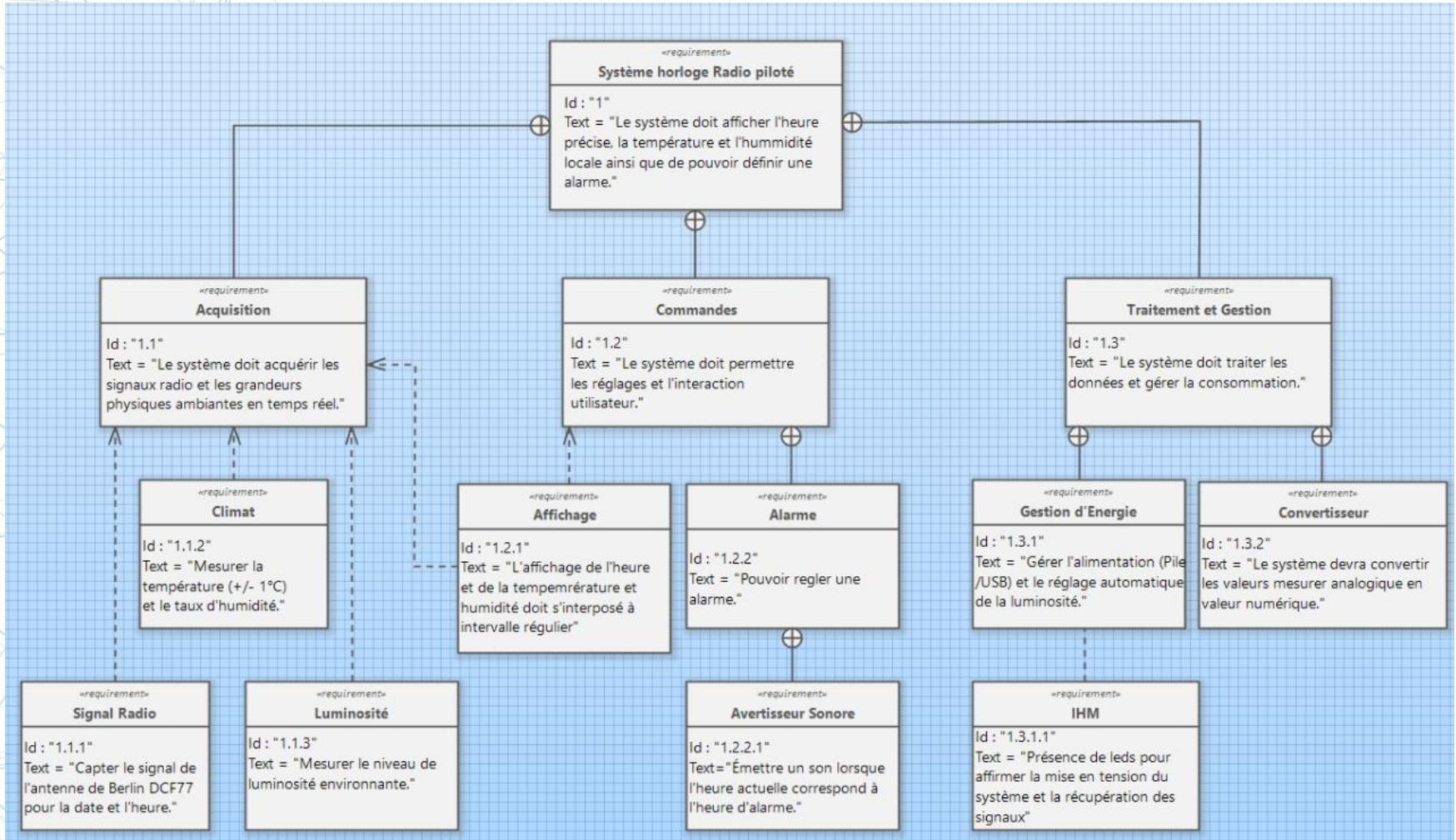


Phase 1: Functional analysis



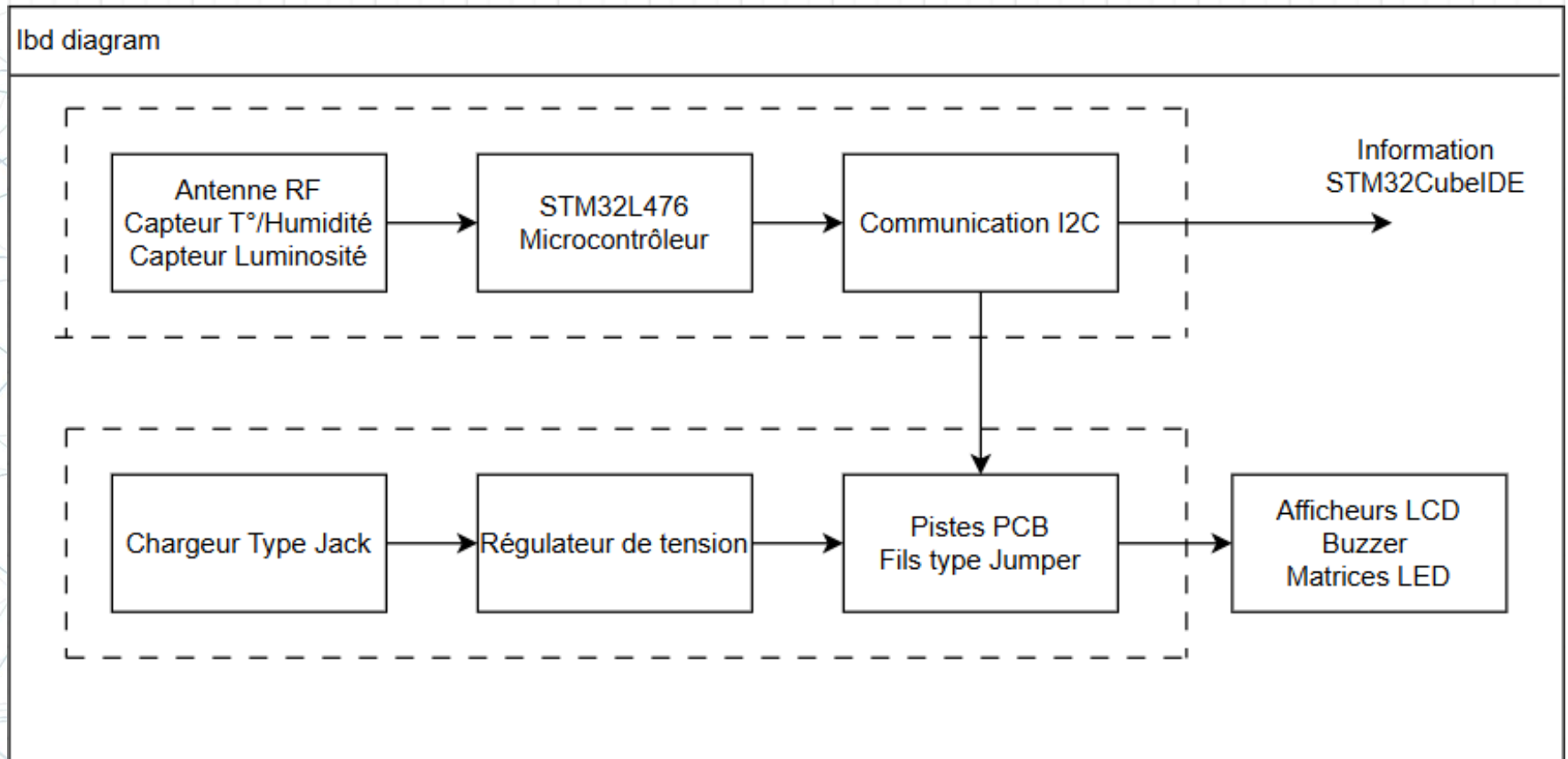
General structure of the project (Ucd)

Specifications diagram



Specifications (simplified by a requirements diagram)

Internal blocs diagram



Flows of energy and data involved

Phase 1: Components Choice

Acquisition:

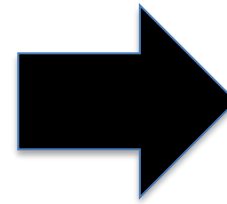
1x SHT41L (Humidity + Temperature sensor)
1x TEMT6000X1 (brightness)
1x DCF77 (RF Antenna)

Power supply:

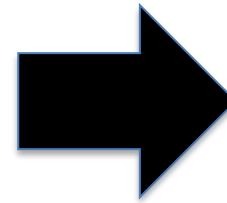
1x BU33D5 (5V-3,3V regulator)
1x Dissipateur (for L78)
1x L78 (12V-5V regulator)
1x Pj-102 plug

display:

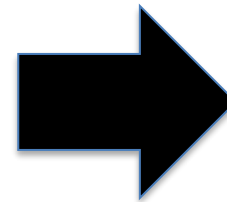
1x Buzzer
1x I2C displayer
5x LED matrice



1 ACQUISITION PCB



1 POWER SUPPLY PCB

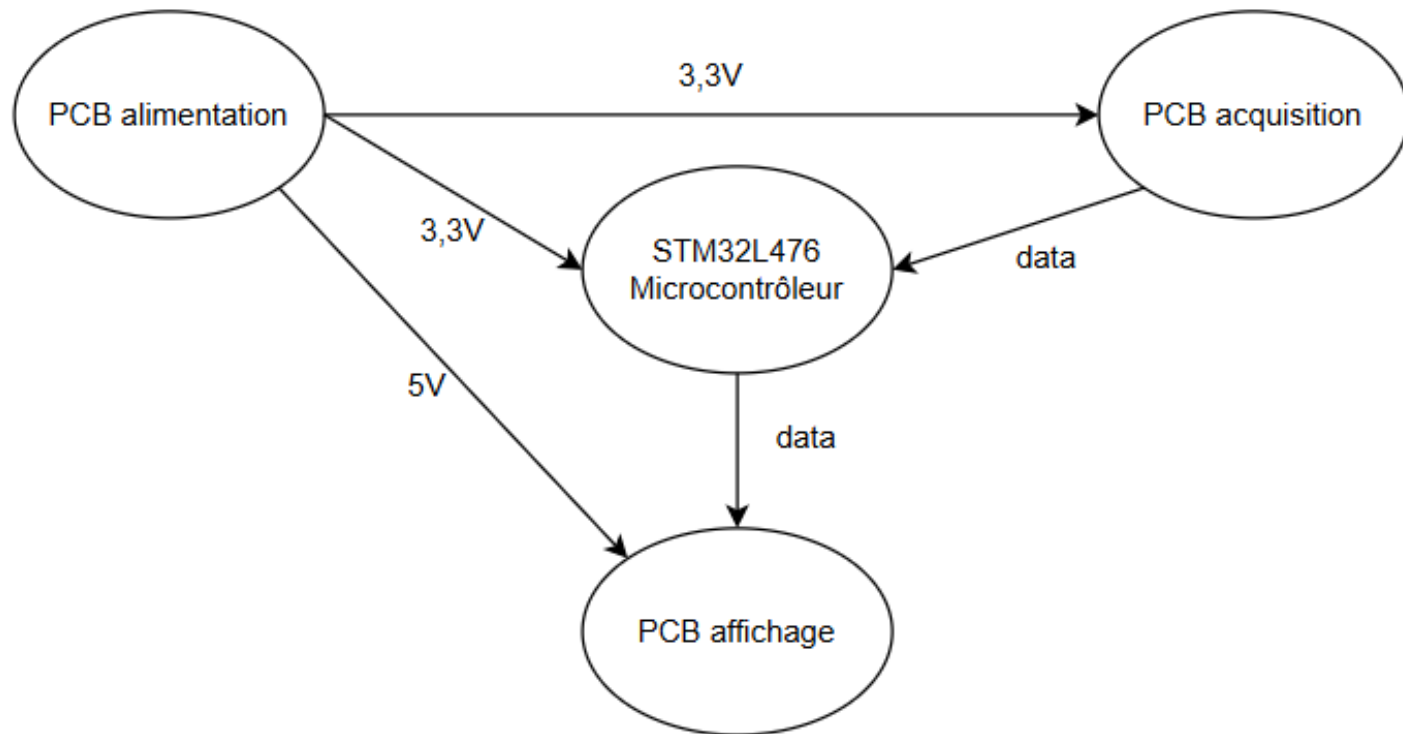


1 DISPLAY PCB

Main suppliers: RS Components and Farnell

Implementation Schematic

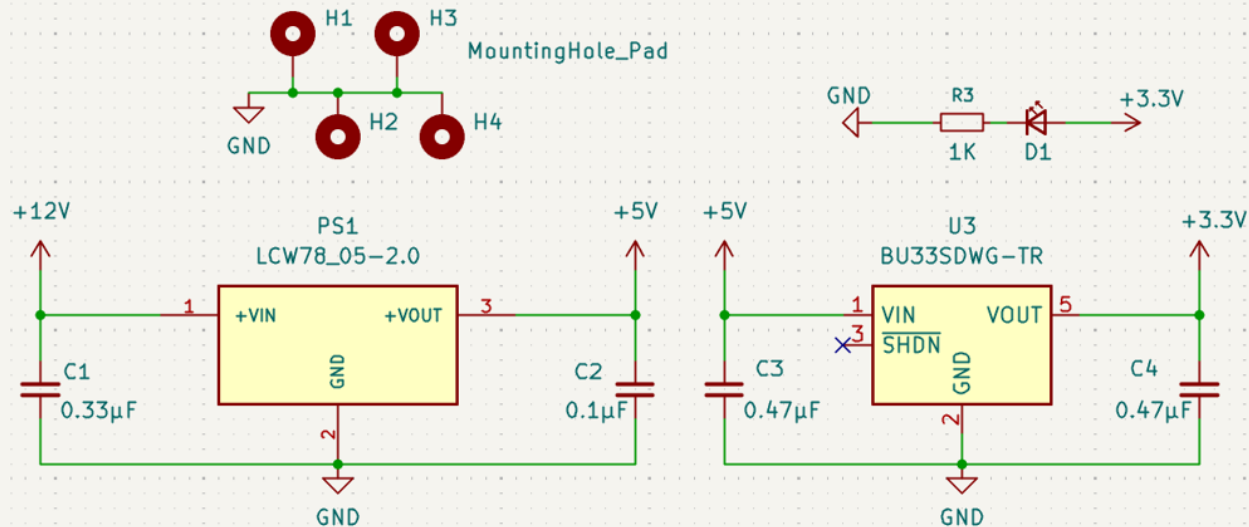
Schéma d'Implémentation



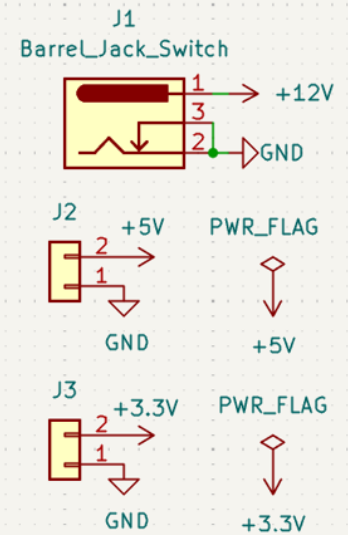
Phase 2: PCB Conception

Alimentation Schematic

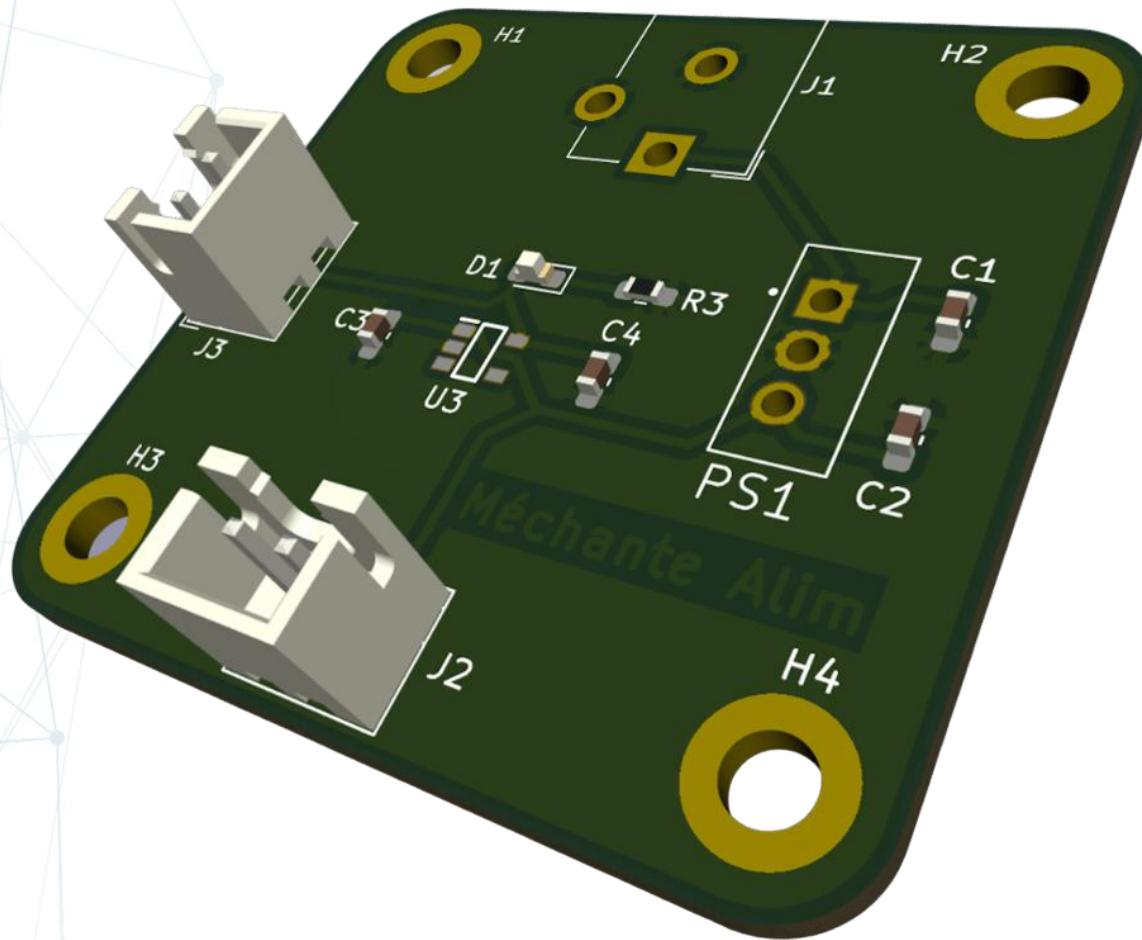
ALIMENTATION et REGULATION DE TENSION



Ports

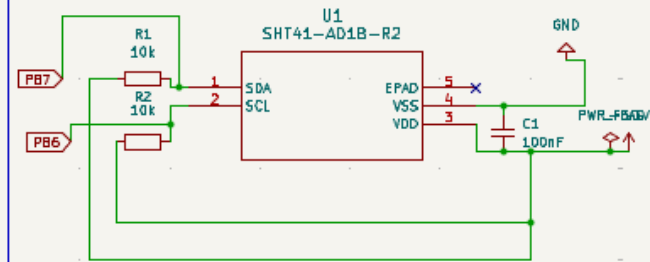


3D View:

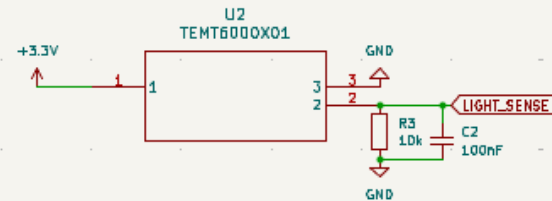


Acquisition Schematic:

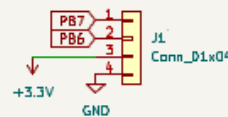
Capteur de température+humidité



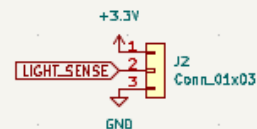
Capteur de luminosité



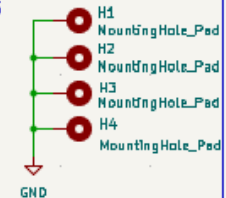
Connecteur SHT40



Connecteur TEMT

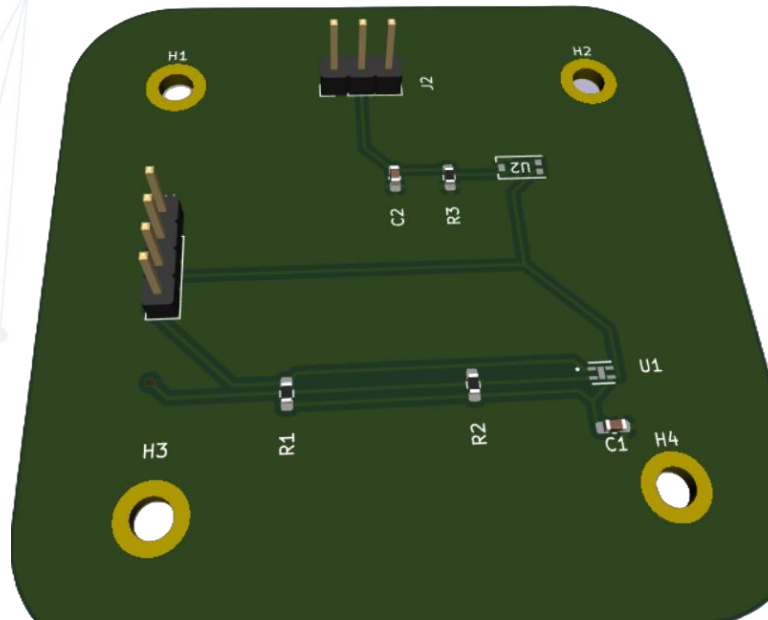


Mounting holes

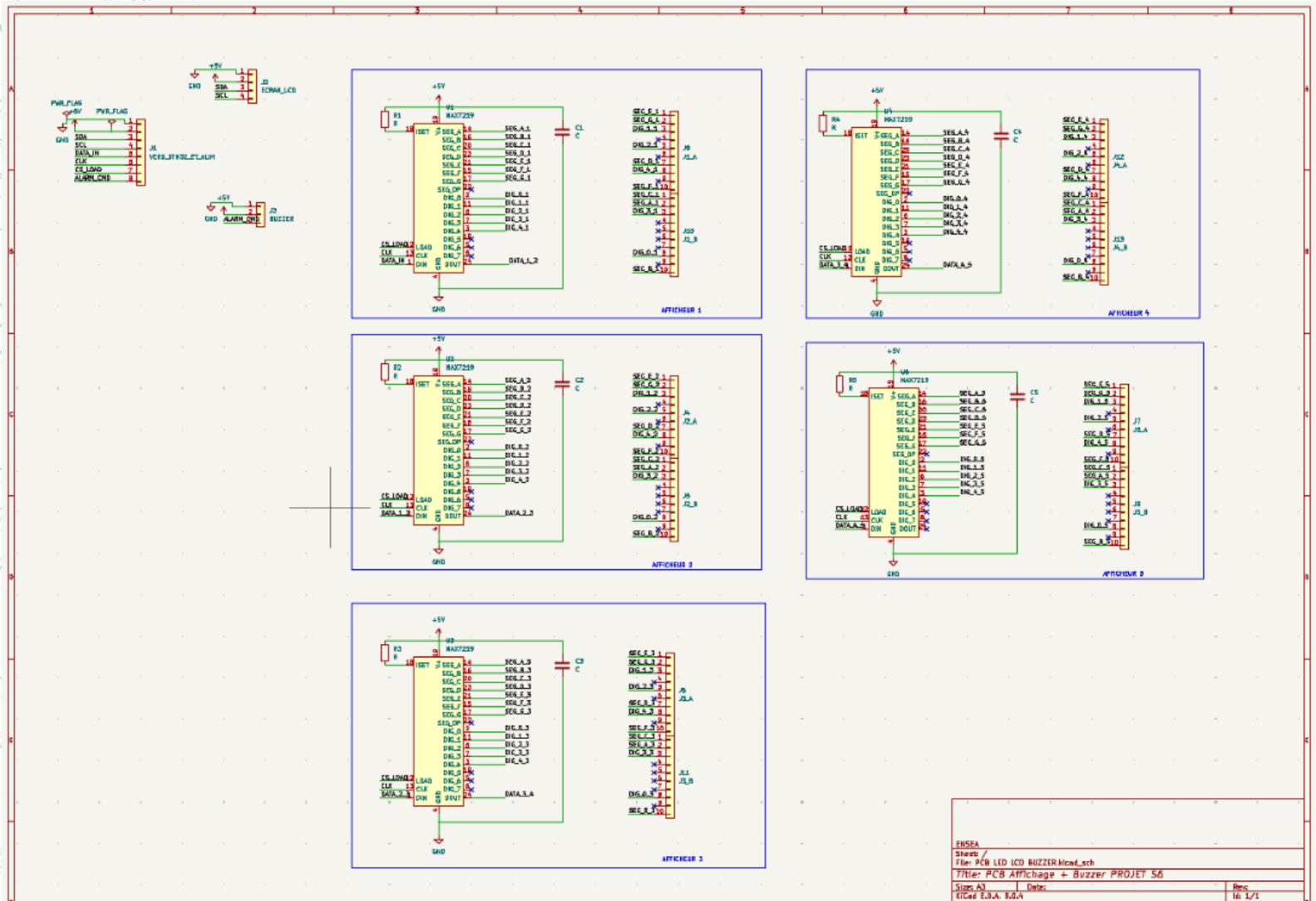


ENSEA
Sheet /
File: Projet_capteur.kicad_sch
Title: Capteurs_PCB

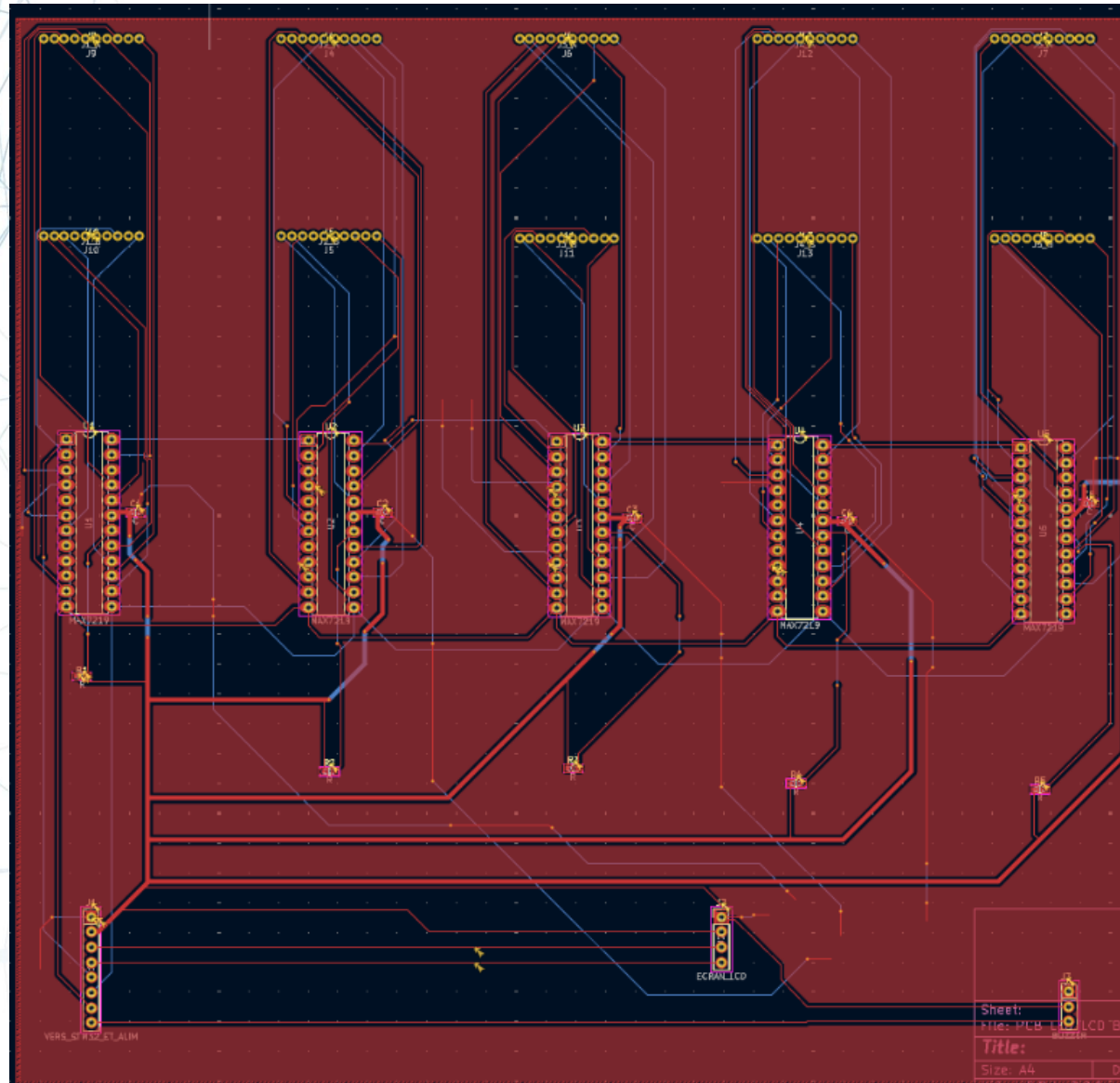
3D View:



PCB Display



3D View:



And now ?

Beginning of the 3rd phase: programming of components

Some bureaucratic tasks (writing project report, github...)

Welding+ mounting

Tests and validation

PCB must have been received after April's vacation